

Chapter 2

Thinking Like an Economist

Learning Objectives

- 2.1 Describe how economists apply the methods of science; explain how models shed light on the world.
- 2.2 Use the circular-flow model and PPF to analyze choices and trade-offs.
- 2.3 Distinguish microeconomics from macroeconomics.
- 2.4 Distinguish positive from normative statements.
- 2.5 Explain economists' roles in policy.

Key Points

- Economists try to address their subject with a scientist's objectivity. Like all scientists, they make appropriate assumptions and build simplified models to understand the world around them. Two simple economic models are the circular-flow diagram and the production possibilities frontier.
- The field of economics is divided into two subfields: microeconomics and macroeconomics. Microeconomists study decision making by households and firms and the interactions among households and firms in the marketplace. Macroeconomists study the forces and trends that affect the economy as a whole.
- A positive statement is an assertion about how the world is. A normative statement is an assertion about how the world ought to be. When economists make normative statements, they are acting more as policy advisers than as scientists.
- Economists who advise policymakers sometimes offer conflicting advice either because of differences in scientific judgments or because of differences in values. At other times, economists are united in the advice they offer, but policymakers may choose to ignore the advice because of the many forces and constraints imposed by the political process.

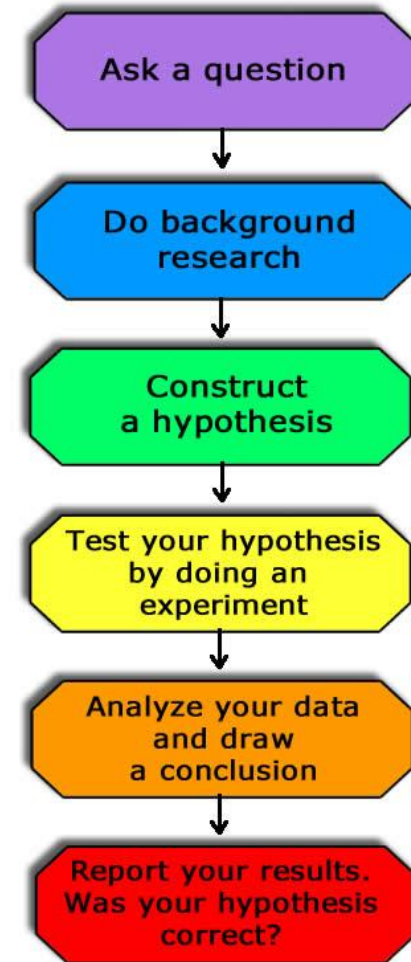
The Economist as Scientist

Economists play two roles:

1. Scientists: try to explain the world
2. Policy advisors: try to improve it

- In the first, economists employ the **scientific method**, development and testing of theories about how the world works
- To use the scientific method, economists make use of models.
- **Model** – a simplification of reality

The Scientific Method



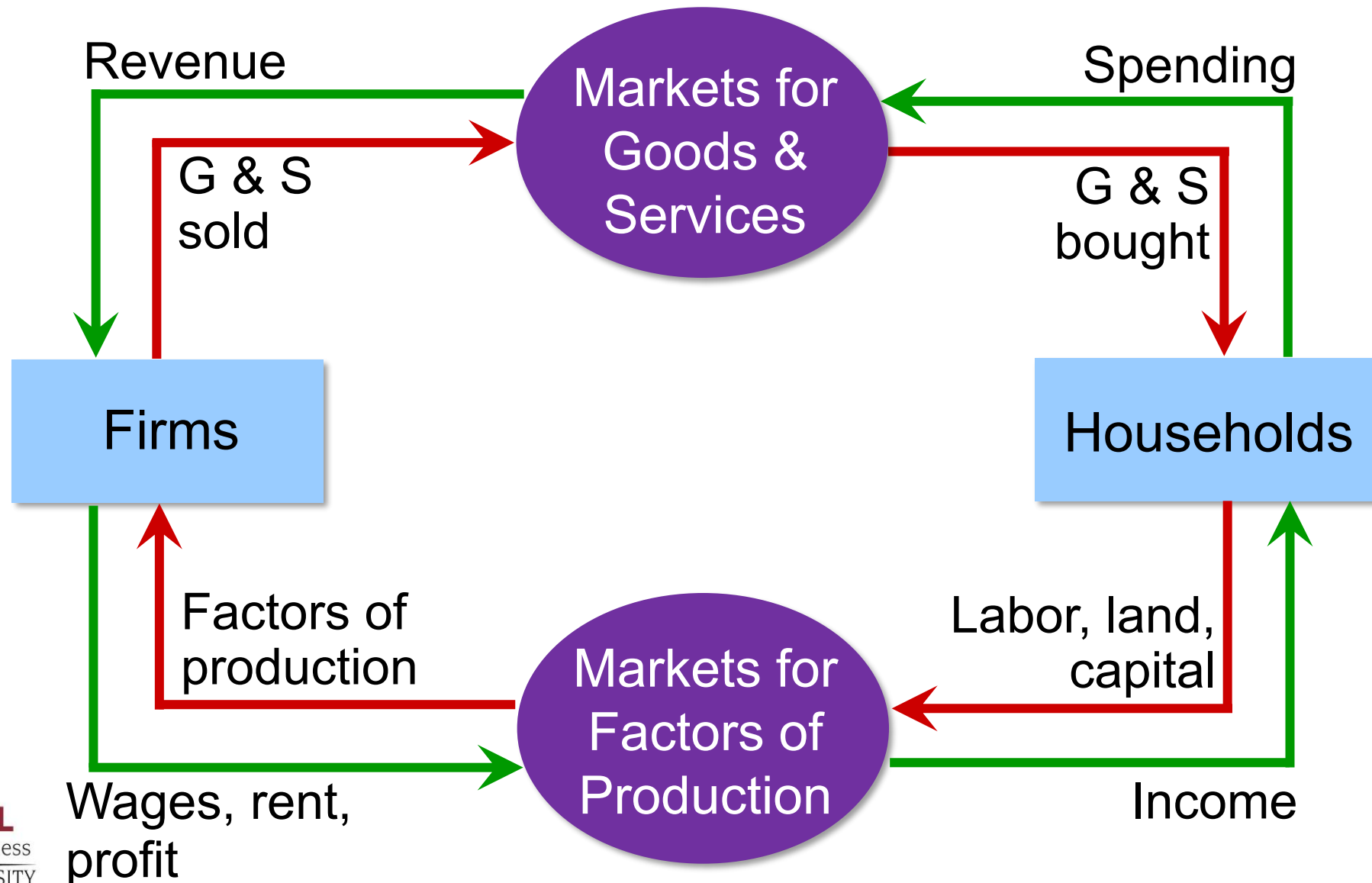
The Circular Flow Diagram

- **Circular-Flow Diagram** - a visual model of the economy, shows how dollars flow through markets among households and firms
- In a simple model, the economy has two agents: Households and Firms and two markets: Product and Resource

Factors of Production “Resources”

1. **Land** – natural resources or “gifts of nature”
 - Rent – payment for the use of land or natural resources
2. **Capital** – buildings, equipment, tools
 - Interest – payment for the use of capital
3. **Labor** – mental and physical abilities
 - Wages – payment to labor for work done
 - a. **Human Capital** – knowledge acquired to do a job more efficiently
4. **Entrepreneurial ability** – special talent that combines other factors of production
 - Profit – payment to entrepreneurs for taking risk and organizing other factors of production.

The Circular Flow Diagram



The Production Possibilities Frontier

- **Production Possibilities Frontier** - a graph that shows the combinations of two goods the economy can possibly produce given the available resources and the available technology

Assumptions of the PPF

- Quantity and quality of labor fixed (full employment)
 - Quantity and quality of capital fixed
 - Technology is in a fixed state
 - Produce only two goods
 - Looking at a specific time
-
- When all these assumptions hold, the PPF creates an *efficient* use of resources to create the level of goods desired

Production Possibilities Table

- The PPF is constructed from a production possibilities table

Possibility	iPhones	DVDs	Loss/Gain
A	60	0	
B	58	15	-2/+15
C	53	27	-5/+12
D	45	37	-8/+10
E	35	45	-10/+8
F	20	50	-15/+5
G	0	52	-20/+2

- Notice that there isn't a 1 to 1 trade off between iPhone production and DVD production.
- 2 iPhones are lost to produce 15 DVDs
- Viewed another way: to gain 1 DVD, 2/15 of an iPhone must be given up
- This is because the resources and technology isn't perfectly adaptable to each good.

Opportunity Cost in the PPF

Possibility	iPhones	Loss/Gain	DVDs
A	60		0
B	58	-2/+15	15
C	53	-5/+12	27
D	45	-8/+10	37
E	35	-10/+8	45
F	20	-15/+5	50
G	0	-20/+2	52

The marginal (additional) opportunity cost of moving from A to B is giving up 2 iPhones.

The total opportunity cost of moving from A to G is giving up 60 iPhones to get 52 DVDs.

When an economy decides to move from one possibility to the other it must give up some of the other good.

Stated differently, in order to get more of one good, a successively larger amount of the other good must be given up. This is called the **Law of Increasing Opportunity Cost**.

This works in reverse as well. Moving from E to D, the marginal opportunity cost is 8 DVDs.

Activity – Opportunity Costs in the PPF

1. What is the marginal opportunity cost of moving from C to D?
2. What is the total opportunity cost of moving from E to B?

Possibility	Beer	Wings
A	0	20
B	2	18
C	4	13
D	9	7
E	13	1
F	15	0

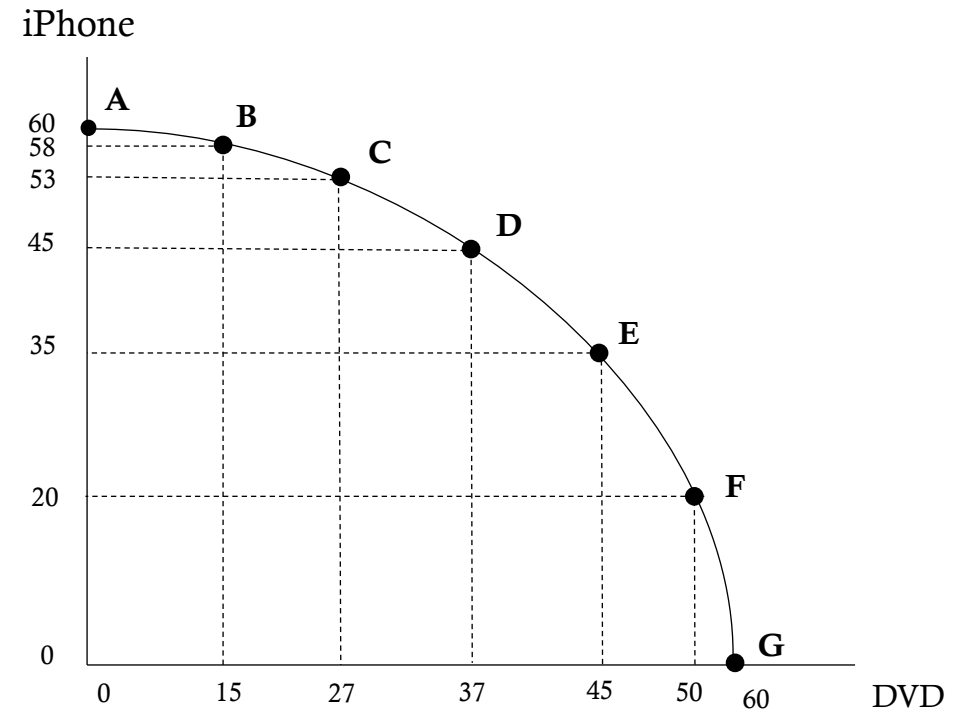
Activity – Opportunity Costs in the PPF

1. Give up 6 wings to get 5 beers
2. Give up 11 beers to gain 17 wings

Production Possibility Curve (PPC)

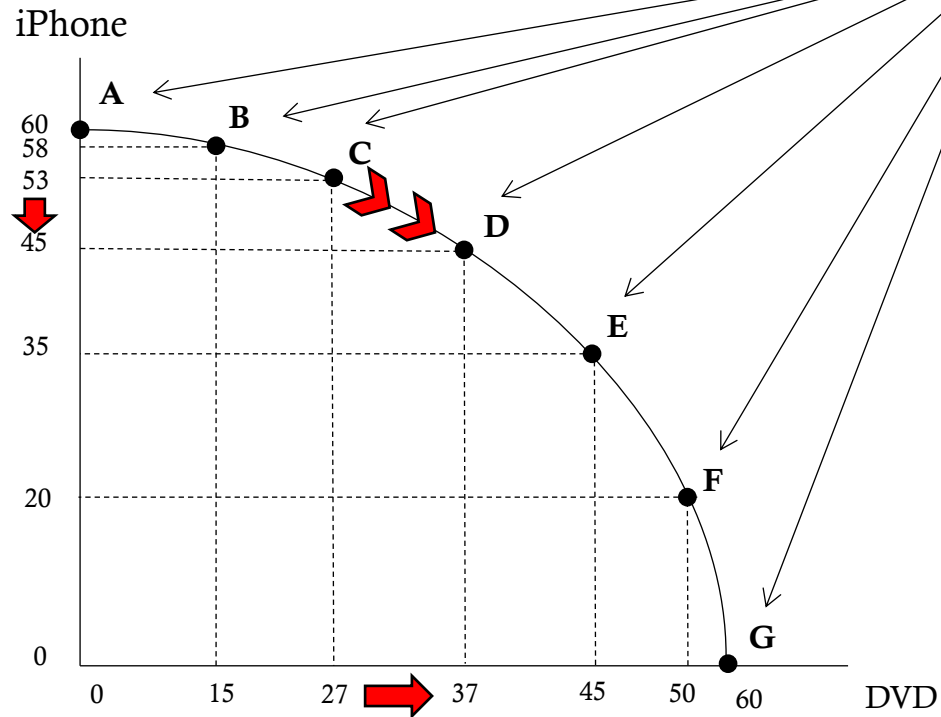
- A production possibilities frontier is graphically illustrated by simply plot the numbers from the production possibilities table onto a graph and thus becomes a production possibility curve (PPC)

Possibility	iPhones	DVDs
A	60	0
B	58	15
C	53	27
D	45	37
E	35	45
F	20	50
G	0	52



The law of increasing opportunity cost causes the PPF to have a **concave** shape (i.e. bowed outward from the origin)

Understanding the PPC



Notice that each possibility sits on the production possibility curve

Why is this the case?

Efficient use of resources

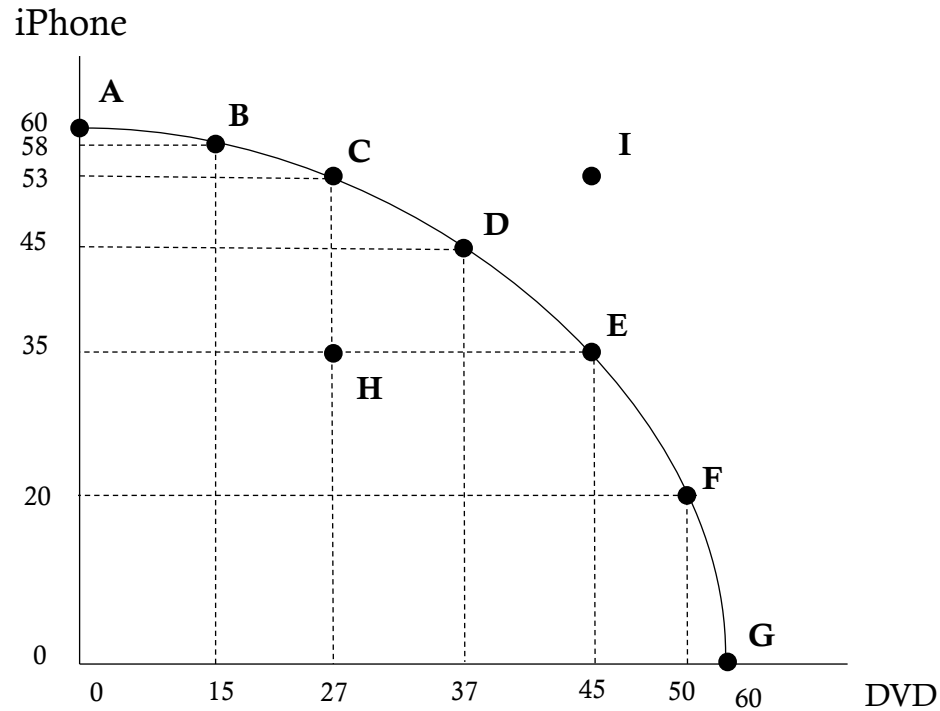
What is the marginal opportunity cost of moving from C to D?

8 iPhones

Which point does the economy choose?

Whichever point equates marginal benefits to marginal costs and gives society what it wants the most.

Efficient, Inefficient, and Attainableness



Points A-G represent production possibilities that are attainable and efficient.

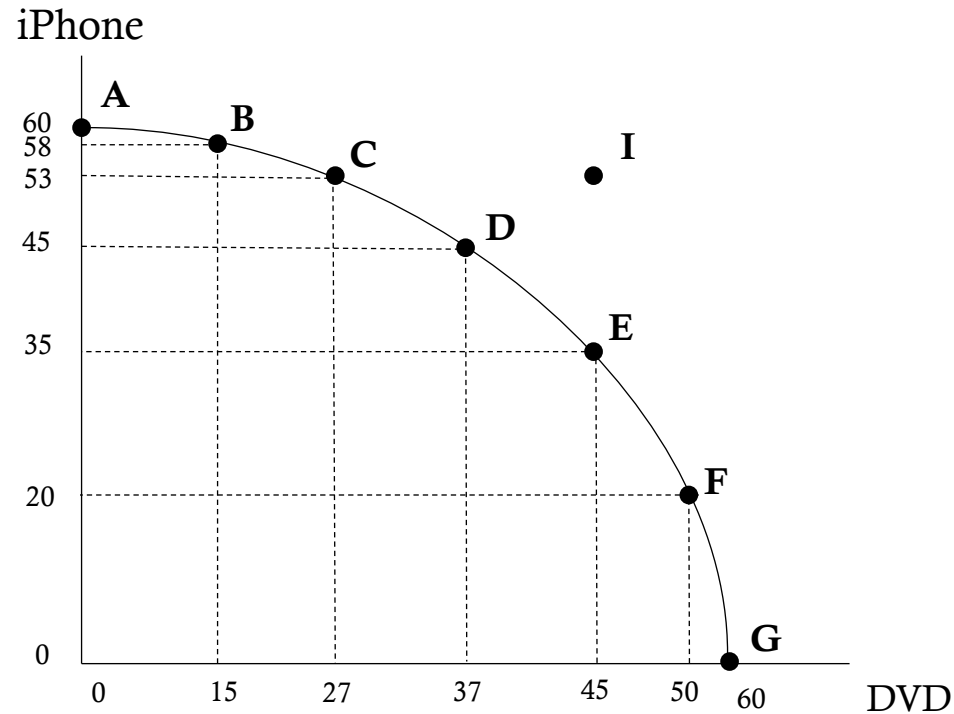
What about point H?

We can certainly get this possibility if we want, but we would be better off if we chose point C (18 extra iPhones) or point E (18 extra DVDs). Point H is therefore attainable but inefficient. It can be best thought of as **unemployment or idle resources**.

What about point I?

Point I lies outside our PPF. In our current state, we do not have enough resources or technology to attain point I. Therefore, point I is **unattainable**.

Expanding the PPC



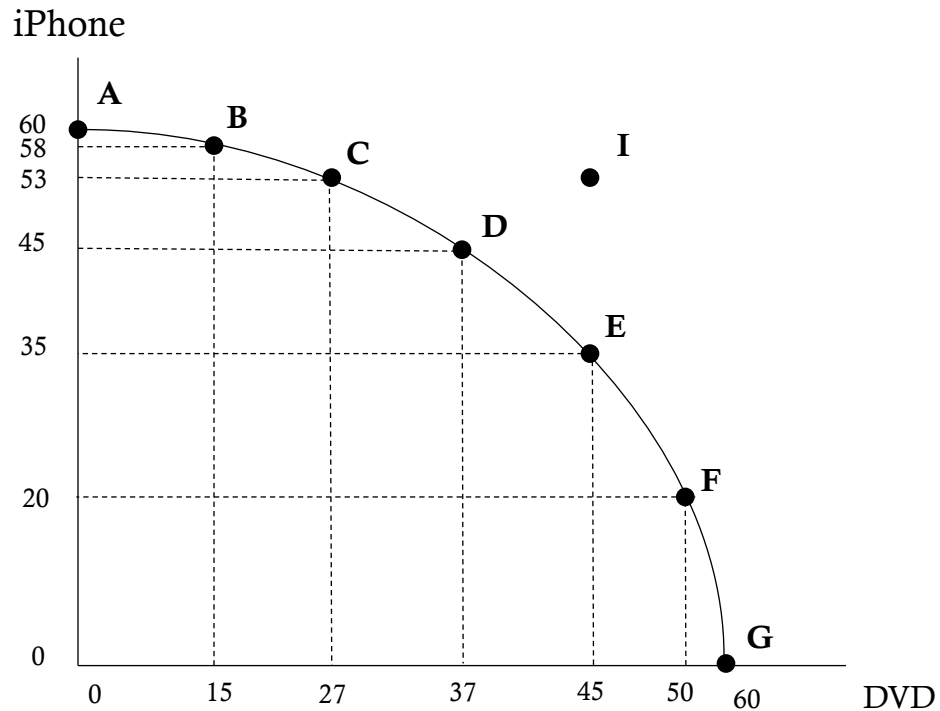
Point I is unattainable in our current state. How do we get to it then?

Recall the assumptions of the PPF:

- Quantity and quality of labor fixed (full employment)
- Quantity and quality of capital fixed
- Technology is in a fixed state
- Produce only two goods
- Looking at a specific time

If we alter the first three assumptions in a positive way, we can shift our PPF towards I.

Expanding the PPC



Shifting the PPF outwards is called **economic growth** and is a result from increasing a factor of production

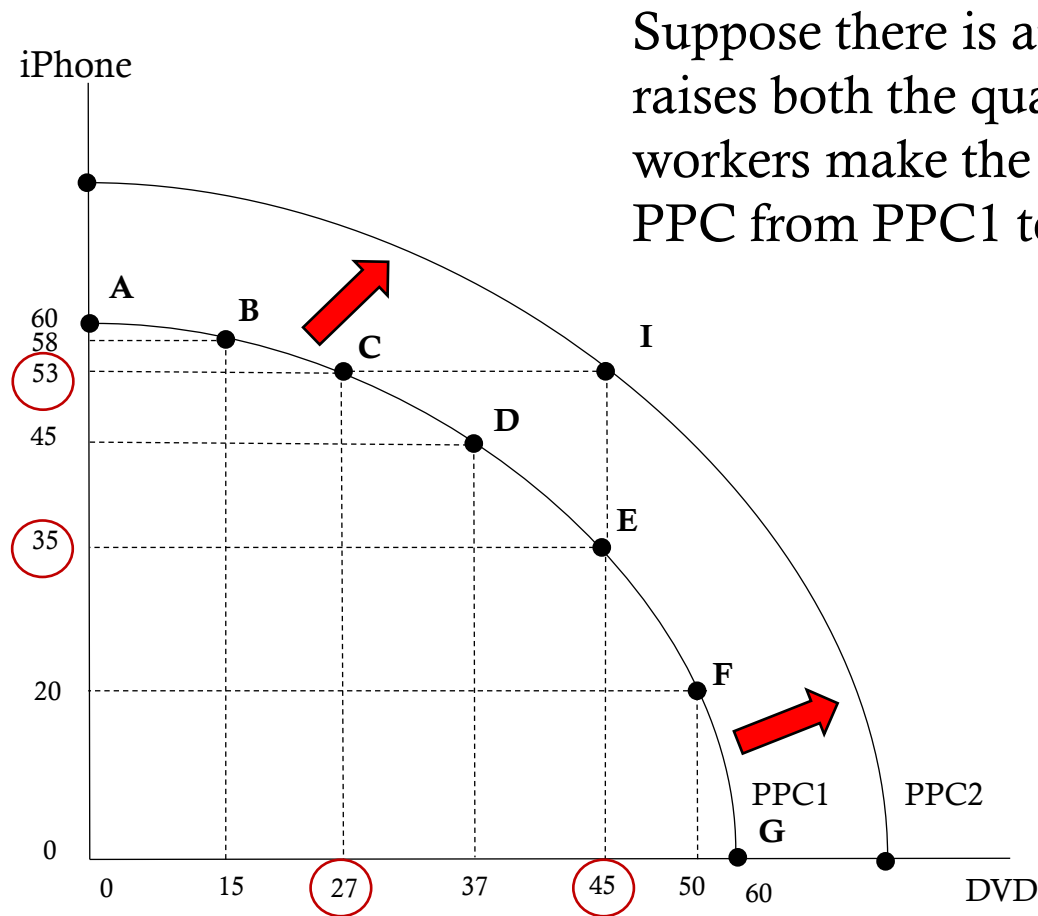
Increasing Factors of Production

1. Increase the quantity/quality of labor.
2. Increase the capital stock in the economy
3. Improve the technological state

We can achieve the above by:

1. Adding more people to the labor force or make them better educated/more skilled
2. Saving more to promote investment
3. Finding new ways to produce a product that causes gains from efficiency

Shifting the PPC (Economic Growth)



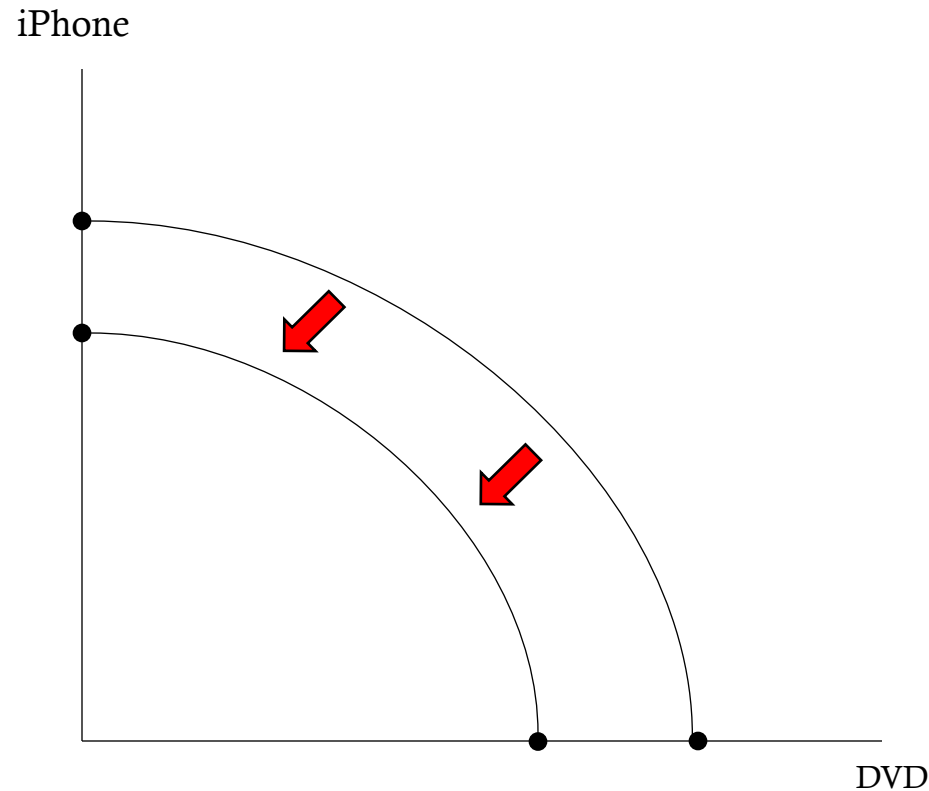
Suppose there is an influx of skilled immigrants to the US. This raises both the quantity and quality of labor. As a result, these workers make the US more productive and cause a shift in the PPC from PPC1 to PPC2.

Compare the new result to the old ones.

1. New (point I) – 53 iPhones and 45 DVDs
2. Old (point C) – 53 iPhones and 27 DVDs
3. Old (point E) – 35 iPhones and 45 DVDs

The result is a gain of 18 iPhones and 18 DVDs

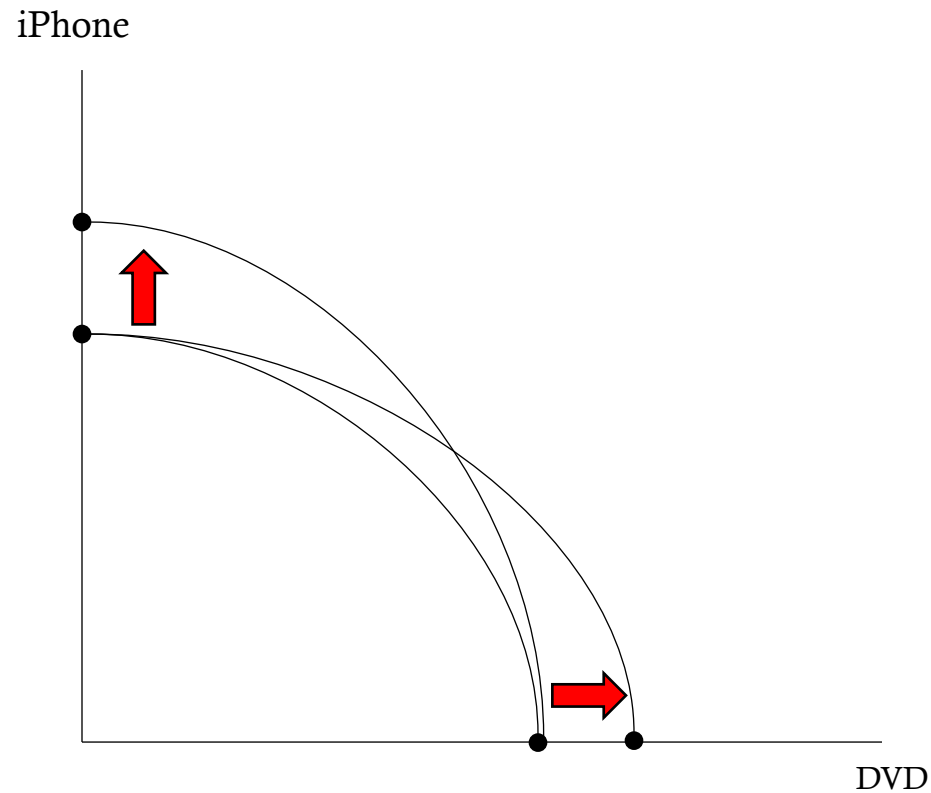
Production Impossibilities?



Economic growth is a shift outwards of the PPF. What about a shift inwards?

This could occur during a natural disasters, e.g. when a hurricane causes production to halt or reverse. Another example is can be found in the movie *Escape from L.A.* where production seems to have completely reversed in Los Angeles.

Benefits to one good and not the other



Suppose a new technology is introduced that makes the production of iPhones more efficient but does not help DVD production. In this case, we would see a rotation of the PPC on the iPhone axis instead of a complete shift.

If the technology improved DVD production, then the rotation would be about the DVD axis instead.

Perfectly Adaptable Resources

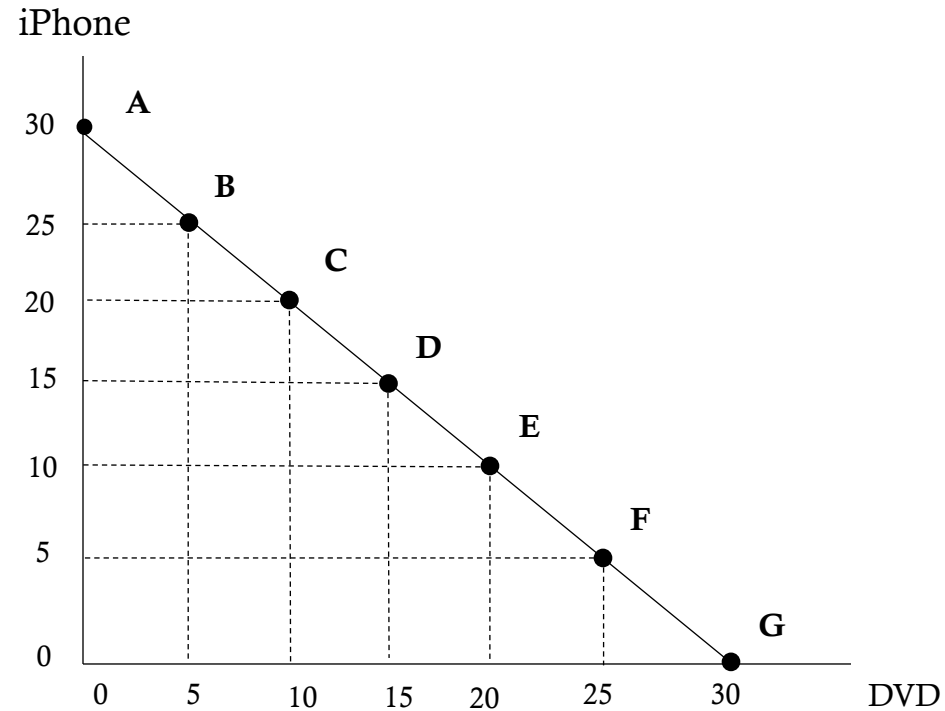
Assume that the technology and resources that are used to produce iPhones and DVDs were perfectly adaptable to one another.

Possibility	iPhones	DVDs	Gain/Loss
A	0	30	
B	5	25	+5/-5
C	10	20	+5/-5
D	15	15	+5/-5
E	20	10	+5/-5
F	25	5	+5/-5
G	30	0	+5/-5

In this case, there is an exact 1 to 1 trade off between iPhones and DVDs. For every iPhone gained one DVD is lost and vice versa.

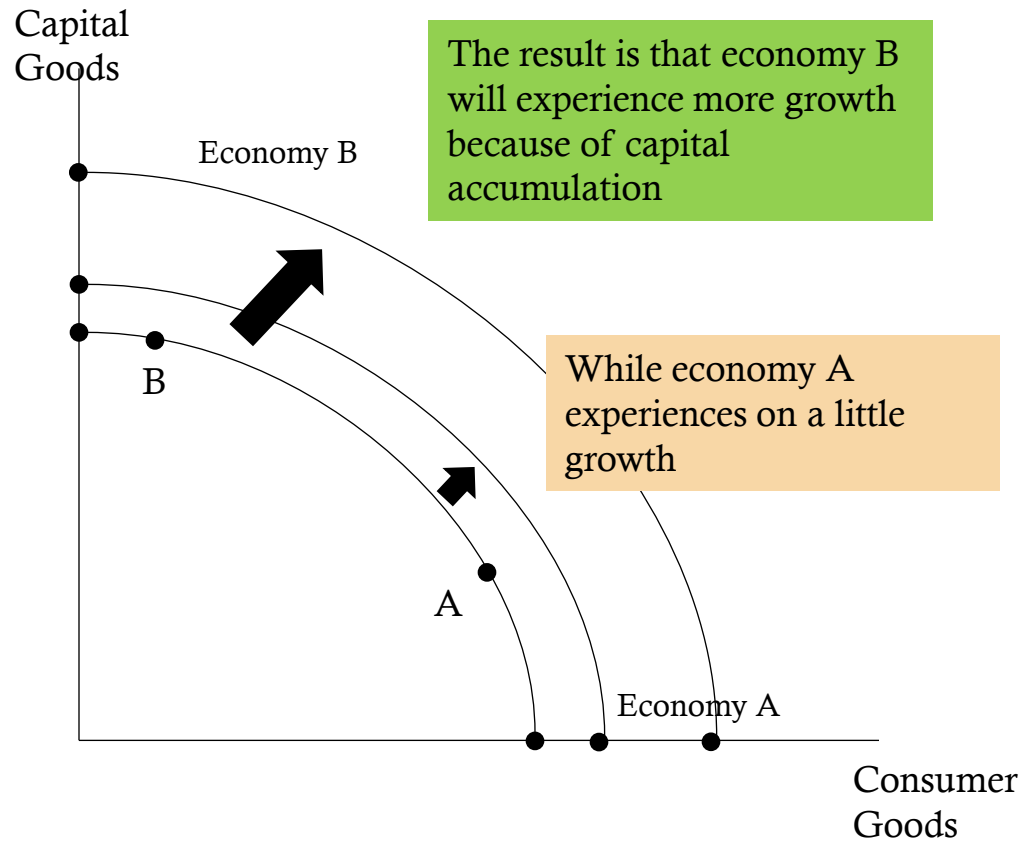
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C	10	20
D	15	15
E	20	10
F	25	5
G	30	0



Due to the 1 to 1 trade off, the PPF is a straight line representing an equal tradeoff between iPhones and DVDs

Current vs. Future Consumption



- Production Possibility Frontiers also help us understand how much economic growth we can expect
- Suppose there are two economies, A & B, that are represented by points A and B on the PPF.
- Economy A prefers current consumption to future consumption indicated by the small amount of capital goods produced and the large amount of consumer goods produced.
- Economy B prefers future consumption over current consumption

Guess the City



Guess the City

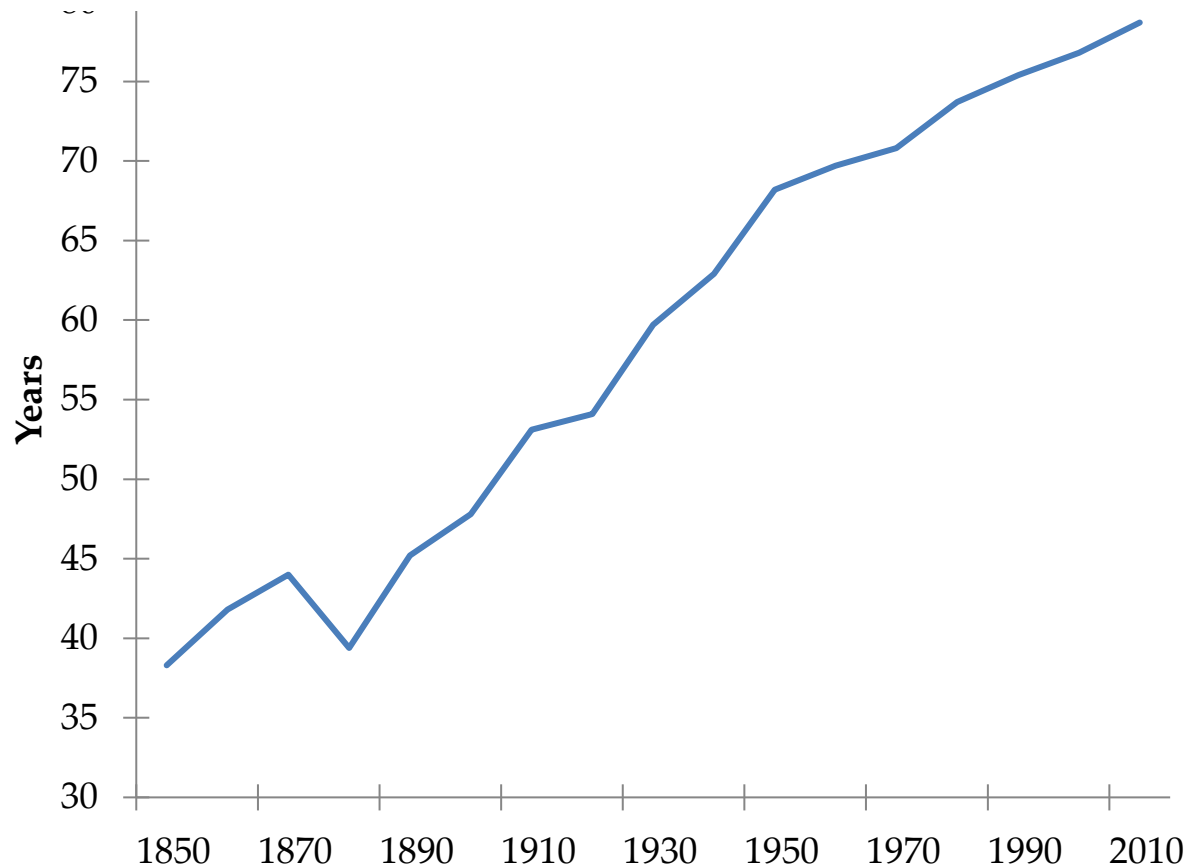


Guess the City

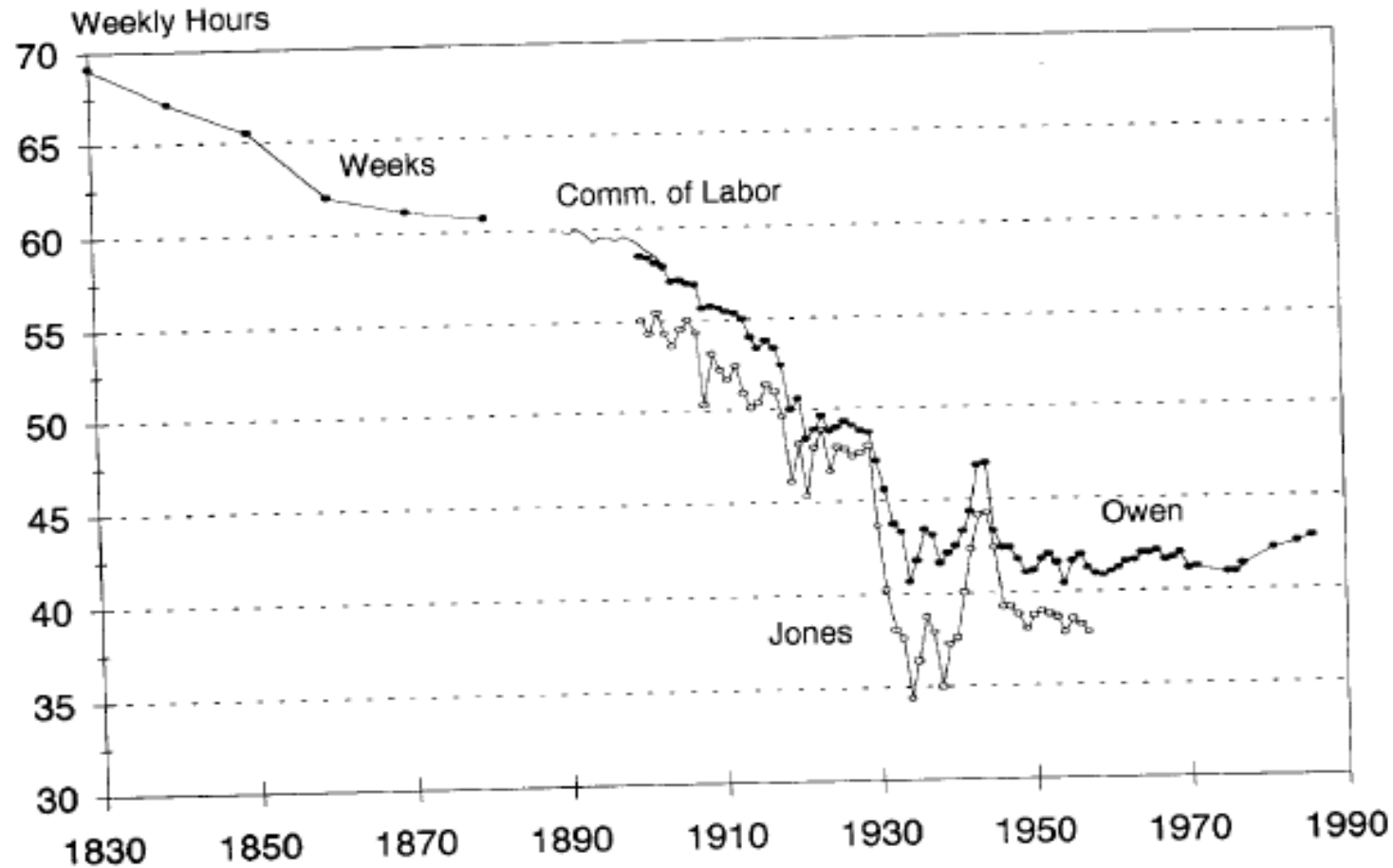


US Historical Life Expectancy

Life Expectancy at Birth, 1850-2010



US Historical Weekly Hours Worked



Source: Golding (1994).

Microeconomics and Macroeconomics

- **Microeconomics** is the study of how households and firms make decisions and how they interact in markets.
- **Macroeconomics** is the study of economy-wide phenomena, including inflation, unemployment, and economic growth.
- These two branches of economics are closely intertwined, yet distinct—they address different questions.

The Economist as Policy Advisor

- As scientists, economists make **positive statements**, which attempt to describe the world as it is.
- As policy advisors, economists make **normative statements**, which attempt to prescribe how the world should be.
- Positive statements can be confirmed or refuted; normative statements cannot.
 - Q: A good way to increase income is to educate more people.
 - A: Normative statement
 - Q: The unemployment rate in December 2017 was 3.7%
 - A: Positive Statement

Why Economists Disagree

- Economists often give conflicting policy advice.
- They sometimes disagree about the validity of alternative positive theories about the world.
- They may have different values and, therefore, different normative views about what policy should try to accomplish.
- Yet, there are many propositions about which most economists agree.

Why Economists Disagree

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Propositions about Which Most Economists Agree (and % who agree)

- A ceiling on rents reduces the quantity and quality of housing available. (93%)
- Tariffs and import quotas usually reduce general economic welfare. (93%)
- The United States should not restrict employers from outsourcing work to foreign countries. (90%)
- The United States should eliminate agriculture subsidies. (85%)

Key Takeaways

- Economists use the **scientific method** to explain how the world works.
- Assumptions and models simplify reality to highlight key ideas.
- **The Circular Flow Model**
 - Shows how households and firms interact in markets.
 - **Factor payments:** rent (land), wages (labor), interest (capital), profit (entrepreneurship).
- **The Production Possibilities Frontier (PPF)**
 - Illustrates trade-offs and opportunity cost between two goods.
 - Points: on = efficient | inside = inefficient | outside = unattainable.
 - Growth shifts the frontier outward.
- **Micro:** decisions of households & firms.
- **Macro:** economy-wide issues—growth, inflation, unemployment.
- **Positive:** describes “what is.”
- **Normative:** suggests “what should be.”

Knowledge Check

1. Why do economists use models like the circular flow diagram?
2. In the circular flow diagram, what flows from households to firms?
3. A point inside the PPF represents:
4. Which of the following is a **macroeconomic** question?
 - a) Why did Starbucks raise its prices this month?
 - b) How many workers should a local factory hire?
 - c) What causes inflation to rise nationwide?
 - d) Should a farmer grow more wheat or corn?
5. Identify the normative statement:
 - a) The unemployment rate is 4.1 percent.
 - b) Higher education increases average income.
 - c) The government should raise the minimum wage.
 - d) Imports account for 15 percent of GDP.

Knowledge Check Solutions

1. To simplify reality and highlight key relationships
2. Factors of production (labor, land, capital)
3. Unused or misallocated resources
4. What causes inflation to rise nationwide?
5. The government should raise the minimum wage